CryoChain: manual testing guide

This walks through what the app is supposed to do, role by role, with the steps and the result you should see. Use it to learn the product and to smoke-test a build. Automated tests are listed at the end.

1. Setup

You need the app running (`npx expo start --web` is the quickest) against a Supabase project with all migrations applied. See the README for env vars.

Test accounts

Role	How to create	How to sign in
Farmer, Buyer	Self-signup in the app	Phone or email, then a 6-digit code
Ops	No invite is possible for the first one. Create an auth user with an email and password in the Supabase dashboard, then set that user's row in `profiles` to `role = 'ops'`	`/ops` with email and password
Driver	Ops invites from **Team** (needs a phone or email)	Phone or email, then a 6-digit code

Phone codes without sending SMS (local Supabase only). `supabase/config.toml` defines fixed test numbers `233200000001` to `233200000005`, all with code `123456`. Type them as `020 000 0001`. Use five numbers for: farmer 1, farmer 2, buyer, driver, and a spare. On a hosted project these only work if you add the same test numbers in the dashboard; otherwise use email codes.

Paystack. Use a Paystack **test** secret key and the test card/mobile money numbers from Paystack's docs. Never run these steps with a live key.

Minimum data before an order can work. Ops needs at least one product with a **buyer price** and a **farmer price** for the grade (Prices and products), at least one collection point, and the farmer needs a listing with stock. A driver must have the route screen open so a location ping exists in the last 30 minutes.

Suggested cast

Open each role in its own browser profile or window so sessions don't clash: `Ops`, `Farmer A`, `Farmer B`, `Buyer`, `Driver`.

2. How the roles fit together

```
Ops sets up products, prices, team   ->   Farmers list stock   ->   Buyer orders and pays
                                                                          |
                  Ops watches the board, clears Decisions     <---  System allocates farmers, builds the
driver route
                                                                          |
                                       Driver collects at farms, delivers   ->   Farmer share payable   ->
Ops pays out
```

Role	Can do	Cannot do
Farmer	Manage own listings, accept or decline order offers, see own earnings, add a mobile money wallet	See buyer prices or other farmers' data

Role	Can do	Cannot do
Buyer	Browse, cart, pay, cancel unpaid orders, book trucks, manage addresses	See farmer prices or payouts
Driver	See own route, share location, mark pickups collected and drop-off delivered, run truck trips	See orders they aren't routed on
Ops	Everything operational: catalogue, prices, farmers, buyers, orders, decisions, payments, payouts, routes, trucks, team, settings	n/a

3. Test scripts

Each step says what to do and what you should see. Anything else is a bug.

A. Sign-in and roles

#	Step	Expected
A1	Open `/`, enter a new test phone, continue	A code screen appears with a resend countdown
A2	Enter a wrong code	"wrong code" style error, stays on the screen
A3	Enter the right code (`123456` for test numbers)	"Who are you?" with only Farmer and Buyer. Note says drivers and ops join by invitation
A4	Choose Farmer, continue	Farmer home, Listings tab. Quit and reopen: you land back on Farmer home (session resumes)
A5	Sign in again with a number that already has a role	Skips role pick, goes straight to home
A6	Open `/ops`, sign in with a farmer's email and any password	"That email and password do not match" or "does not have operations access", and you are signed out
A7	Open `/ops`, sign in as ops	Ops dashboard with the side menu
A8	Sign out from each role	Returns to landing (ops returns to `/ops`)

B. Ops set-up (do this first)

#	Step	Expected
B1	Team: invite a driver with name, a valid Ghana phone and email	Driver appears under Drivers. Bad phone or email shows an inline error and Send stays disabled
B2	Prices and products: add a category, a product, a buyer price and a farmer price per grade	Product shows in the catalogue; buyers can see it in Shop
B3	Settings: set min and max temperature, min above max is rejected	"Must be above min" and Save is disabled
B4	Settings: save payout threshold, bank details, support phone	Button shows "Saved"
B5	Cold trucks: set GH₵ per km	Saved; used in buyer truck quotes

C. Farmer

#	Step	Expected
C1	Profile: fill name, address (map or fields) and save	"Wallet needed" badge in the header until a wallet exists
C2	Payments: add a mobile money wallet (network, number, name)	Wallet saved after verification; badge disappears
C3	Listings > New listing: pick product, grade, weight, add a photo, pick a collection point, add	Listing card shows Sold 0, Delivered 0, Left = weight
C4	Edit a listing	Changes saved
C5	Remove a listing that has no sales	Removed. Remove is hidden once anything is sold or withdrawn
C6	Look for any buyer price anywhere in the farmer app	None. Only the farmer price and payout amounts

D. Buyer shopping and payment

#	Step	Expected
D1	Account: add a delivery address (map pin or typed) and organization details	Account tab alert dot clears once the profile is complete
D2	Shop: open a product, set kg with the stepper	Cannot go above available stock
D3	Add to cart	Cart tab shows an alert dot
D4	Cart: no address selected	"Check availability & pay" is disabled; "Add address" offered
D5	Cart: kg greater than stock (change stock in another window to force it) then pay	Shortfall panel lists what is available, with "Continue with available" and "Review my cart". Per line "Reduce to N kg" also works
D6	Cart with enough stock: Check availability & pay	Order created `awaiting_payment`, stock held for 30 minutes, payment page opens
D7	Pay with a Paystack test method	Order becomes **paid**. Orders tab shows Order placed then Payment received
D8	Abandon payment, return	Orders tab shows "Pay now" / "Try payment again" and "Cancel order"
D9	Cancel an unpaid order	Order shows Cancelled; the stock is available again
D10	Wait over 30 minutes on an unpaid order, then try to pay	The order is cancelled instead of charged
D11	Look for any farmer price in the buyer app	None

E. Allocation and farmer offers (runs automatically after D7)

#	Step	Expected
E1	Farmer A (the nearest to the buyer): Messages tab	"Orders for your produce" shows an offer with kg and payout amount, Accept and Decline
E2	Farmer A accepts	Offer becomes accepted; ops Orders board shows the allocation as accepted
E3	Another order: Farmer A declines	Offer is re-allocated to the next farmer (Farmer B) with a new offer
E4	Another order: nobody responds	After the 2 hour window an expiry job (every 10 minutes, needs pg_cron) marks it expired and re-allocates
E5	Order bigger than any single farmer's stock	Split across more than one farmer

#	Step	Expected
E6	Order for a product with no total supply	Ops **Decisions** shows "Not enough supply" with that order
E7	Product with no farmer price	Decisions shows "No farmer price"
E8	SMS provider off or failing	Decisions shows "SMS not delivered". Farmer can still accept in the app
E9	Fix the cause, then Decisions > Retry allocation	Allocation runs again; the decision auto-resolves or can be resolved with a note of 3+ characters

F. Driver route and delivery

#	Step	Expected
F1	Driver opens the app and allows location	Today's route shown. If denied: "Allow location so we can send you the nearest orders"
F2	Paid, allocated order exists	Within a moment the route lists farm pickups first, then the buyer drop-off, numbered, nearest-first
F3	Open Navigate on a stop	Opens Google or Apple Maps to that point
F4	Tap Collected on a pickup	Stop moves to done; allocation becomes collected
F5	Try to deliver before all pickups are done	Drop-off is never ordered before its pickups
F6	Tap Delivered on the drop-off	Order complete; each farmer's allocation payout becomes **payable** and the farmer gets an "Order delivered" notification
F7	No driver has pinged in the last 30 minutes, order paid	Decisions shows "No driver available". Once a driver opens the app the route builds itself
F8	Two orders, two drivers	The new order goes to the driver with the lighter load

G. Payouts (Ops and Farmer)

#	Step	Expected
G1	Farmer: Payments tab before delivery	Earnings show as pending
G2	Ops, Farmer payouts	Delivered orders appear under "To pay". Undelivered ones wait under "Waiting" until delivery
G3	Select payable items (or Select all payable), create a run	Run is created and pre-checked
G4	Run total above the payout threshold	Needs a second ops user to approve before release
G5	Release funds	Paystack transfers are sent; items move to Sent, then Paid
G6	A transfer fails	Use Retry; a sent transfer can be checked with Check with Paystack
G7	Farmer Payments tab	Delivered and paid amounts reflect the above
G8	Farmer without a wallet	Cannot be included in a run

H. Cold truck booking

#	Step	Expected
H1	Buyer, Truck: choose Transfer or Pick up, set From and To (map, search or "use my location"), pick a time	A quote appears from the per-km rate

#	Step	Expected
H2	Time in the past (more than 10 minutes ago)	Book is disabled
H3	Book this truck	Booking `awaiting_payment`, listed under Your trips
H4	Pay for this truck	Booking becomes **paid**
H5	Ops Cold trucks: Assign a driver; optionally Set fare, Confirm payment, or Cancel booking	Driver sees the trip under Cold truck trips
H6	Buyer opens a paid trip before it starts	Sees a 4-digit start code
H7	Driver enters a wrong code, then the right code	Wrong rejected; right starts the trip ("On the way")
H8	After start	Buyer sees the truck moving on the map; both can use trip chat
H9	Cancel an unpaid booking	Cancelled; any pending payment is marked abandoned

I. Ops oversight

#	Step	Expected
I1	Dashboard	Counts reflect orders, supply, decisions, payments, payouts; panels link through
I2	Orders board	Each order shows buyer, amount, items, allocations with farmer, kg and status, and the driver's stops in order. "Re-run allocation" works
I3	Payments in	Buyer payments listed; stuck ones visible for review
I4	Supply, Farmers, Buyers	Lists match what farmers and buyers created
I5	Routes	Per-driver stops and states
I6	Team	Invited staff listed by role
I7	Non-ops user calls ops data directly (try with a farmer's session)	Rejected by row level security; no rows returned

4. Edge cases worth trying

- Two buyers racing for the last kg: only one should hold it.
- Pay the same order twice (double tap, or reload on the payment page): charged once.
- Paystack webhook arriving before the browser returns: order still ends up paid exactly once.
- Amount or currency mismatch from Paystack: payment is flagged `mismatch`, order not marked paid.
- Reload on a deep web URL: returns to landing unless a session is saved.
- Offline: actions show an error banner rather than silently failing.
- Dark mode toggle: every screen stays readable.
- Narrow phone vs wide desktop: ops uses a side menu on wide screens; farmer and buyer use bottom tabs.

5. Automated tests

```
npm test               # Jest: auth service, cart logic, country data, shared components
npm run test:functions # Deno: Paystack signature check
```

```
npx tsc --noEmit
npx expo lint
```

There are no end-to-end tests. Sections C to H above are the regression script until some exist.

6. Not built yet (don't file as bugs)

From the client meeting in `docs/Flow.md`: partial-fulfilment confirmation after payment, refunds and refund approval, dispute workflow with payout holds, driver evidence capture (quantity and quality at pickup), live truck map for product orders (live tracking exists for truck bookings), route deviation alerts, and ETA.